

Jingyu Fu

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EDUCATION

Southern University of Science and Technology (SUSTech) Sep. 2023 – Jun. 2027 (Expected)

Department of Biomedical Engineering | Major in Intelligent Medical Engineering

Cumulative GPA: 3.81/4.00 **Annual GPA:** 3.92/4.00 **Overall Evaluation Ranking:** 3 / 257

Selected Coursework: Machine Learning, Medical Image Processing, Medical Big Data, Signals and Systems, Data Structures and Algorithms, Medical Imaging Systems, Intelligent Sensor Technology, Neuroengineering and BCI, Medical Robotics, Probability Theory, Linear Algebra

Research Interests: Medical Image Analysis, Computer Vision

PUBLICATIONS & PATENTS

Journal & Conference Papers

1. Q. Wei, **Jingyu Fu**, et al., “Smartphone Dual-Camera Based Blood Pressure Screening Using Pulse Transit Time between Ear and Finger,” in *Proc. 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2025. DOI: [10.1109/EMBC58623.2025.11252747](https://doi.org/10.1109/EMBC58623.2025.11252747)
2. **Jingyu Fu** (First Author), et al. *Enhancing Thyroid Ultrasound Diagnosis under Severe Class Imbalance ...*. Targeting *Biomedical Signal Processing and Control*, **Manuscript in Submission**.
3. L. Shang[†], Y. Huang[†], X. Feng, **Jingyu Fu**, et al. *A Pathology Foundation Model-Based Dual-Probability Framework for Risk Stratification and Clinical Triage of Cytologically Indeterminate Thyroid Nodules*. **Modern Pathology** (Under Major Revision, Ref: MODPATH-D-26-00161), 2026.

[†] These authors contributed equally to this work.

Patents

1. **Jingyu Fu** (First Student Inventor), Wenjin Wang, et al., *A Non-contact SpO₂ Monitoring Method and System Based on Smartphone DC Calibration*, Chinese National Invention Patent, No. 202511080860.5, **Granted**.
2. **Jingyu Fu** (First Student Inventor), et al., *A Deep Learning-based Framework for Automated Thyroid Nodule Detection and Classification in Ultrasound Images*, Chinese National Invention Patent, **Under Review**.

RESEARCH EXPERIENCE

Topic A: Non-contact Physiological Sensing (PPG / BP / SpO₂)

Smartphone Dual-Camera Blood Pressure Screening via PTT Aug. 2024 – Feb. 2025

Contactless Healthcare Lab, SUSTech — Advisor: Prof. Wenjin Wang / Role: Experimental Lead

- Developed a smartphone-based BP screening pipeline using **ear-finger Pulse Transit Time (PTT)** extracted from synchronized dual-camera rPPG signals, and implemented multi-sensor signal alignment and feature engineering.
- Led the **human-subject experiment design** and data acquisition using an ice-water stimulation protocol; achieved MAE of **3.74 / 2.73 mmHg** (SBP/DBP), meeting AAMI/BHS standards, with results published in **IEEE EMBC 2025** (Second Author).

Multispectral Facial Perfusion and Microcirculation Separation

Oct. 2024 – Feb. 2025

Collab. with Jožef Stefan Institute, Slovenia (Role: Independent Researcher)

- Investigated the separation of **facial macro-perfusion and microcirculation signals** from multispectral RGB videos through spatial-temporal projection.
- Analyzed the feasibility and limitations of depth-resolved physiological signal decoupling under RGB spectral constraints, with a focus on signal coupling across tissue layers.

Non-contact SpO₂ Monitoring via AC–DC Calibration

Feb. 2025 – Jun. 2025

Contactless Healthcare Lab, SUSTech (Role: Primary Inventor)

- Proposed a personalized **AC-DC calibration** strategy using contact PPG (rear camera) to calibrate front-camera DC signals, enhancing robustness against motion and illumination variations.
- Validated the mapping between RoR_{DC} and blood oxygen levels via a breath-holding protocol; filed a **Chinese National Invention Patent** (First Student Inventor).

Topic B: Deep Learning in Thyroid Oncology

AI-Driven Ultrasound Thyroid Nodule Detection & Classification

Sep. 2025 – Present

Collab. with ADA Lab, Hunan University (Role: Independent Researcher)

- Developed a diagnostic pipeline integrating **YOLOv12m** and **SimCLR-based self-supervised learning** to extract stable features from 47k+ unlabeled frames; proposed a **Dynamic Fusion Strategy** (confidence-rule and weighted fusion) to enhance malignant recall to **90.6%**.
- Implemented **DeepSort tracking** and temporal aggregation for patient-level malignancy prediction; built a **Flask+React** platform for video-level visualization and clinical review; filed a **National Invention Patent** and **First-author** manuscript *Targeting Biomedical Signal Processing and Control(BSPC)*.

Foundation Model for Thyroid FNA Cytology Risk Stratification

Sep. 2025 – Feb. 2026

Collab. with Hangzhou Institute of Medicine, CAS & Hunan University (Role: AI Model Developer)

- Contributed to an AI framework for **risk stratification of indeterminate thyroid nodules (Bethesda III–IV)** using pathology foundation-model features extracted by **TITAN**, and participated in the design and implementation of **DBD-Net** for interpretable 2D risk assessment.
- Achieved **AUC 0.948** and **97.9% correct stratification**; manuscript under major revision at *Modern Pathology* (2026).

INTERNSHIP & PROFESSIONAL EXPERIENCE

Research Intern | Pervasive Interaction Lab, Tsinghua University

July 2025 – Nov. 2025

Project: Continuous NIBP Monitoring via Multi-sensor PTT (Mentor: Jiankai Tang)

Beijing, China

- **System Design:** Built an NIBP pipeline via 4-point PPG sensors; synchronized heterogeneous signals (Biopac 2000Hz vs. HUB 111Hz) with time-frequency QC (HR error < 5 BPM). [[PI_Lab](#)]
- **Deployment:** Reproduced the *RingTool* framework and implemented **ONNX-based model deployment** for cross-platform inference; managed remote computing on private cloud servers. [[RingTool](#)]

Clinical Observation Trainee | SUSTech Hospital & Guangzhou University Shenzhen Hospital

July 2024

- Completed a 5-day clinical observation program across departments including trauma, spine surgery, and dental clinics, gaining first-hand exposure to clinical workflows and diagnostic procedures.
- Developed an initial understanding of real-world clinical needs and challenges, informing later research in medical AI system design and applicability.

COMPETITIONS & RESEARCH GRANTS

Intelligent Diagnosis of Lower Limb Atherosclerosis via RGB Camera

2024 – 2025

Core Member / Clinical Collab: Shenzhen Third People's Hospital

- Developed a non-contact diagnostic device for **Peripheral Artery Disease (PAD)** using rPPG-based plantar perfusion imaging and multi-dimensional physiological analysis.
- **Awards & Grants:**
 - **Provincial Third Prize**, "Challenge Cup" Guangdong Undergraduate Academic Science and Technology Works Competition.
 - **Provincial Funding**, Guangdong "Climbing Program" (Special Fund for Scientific and Technological Innovation Strategy).
 - **Provincial Nomination**, National Undergraduate Training Program for Innovation and Entrepreneurship.
- **Technical Contribution:** Designed a multi-modal diagnostic model integrating **Perfusion Index (PI)**, **Pulse Arrival Time Difference (PATD)**, and **Waveform Offset** to classify PAD severity with over 90% accuracy.

- Bayesian Voting System Analysis for Reality Shows (MCM/ICM)

Jan. 2026 – Feb. 2026

Team Member / MCM (Results Pending)

- Developed a **Bayesian Inverse Framework** to infer latent audience preferences from observed rankings and judges' scores; utilized **counterfactual simulation** to analyze the impact of voting rules on system fairness.
- NIPT Screening Strategy Optimization Model (CUMCM)

Sep. 2025

Team Member / **Provincial First Prize**

- Built a statistical model to analyze the correlation between cfDNA concentration, gestational age, and maternal BMI; optimized individualized testing time-points to minimize detection failure rates.
- Olympic Medal Prediction via Machine Learning (MCM/ICM)

Feb. 2025

Team Member / **Meritorious Winner (M Prize)**

- Engineered a predictive framework using **Random Forest and Logistic Regression** to analyze the impact of economic, demographic, and host-country factors on Olympic medal distribution.

TECHNICAL SKILLS

Frameworks & Computer Vision	PyTorch, TensorFlow, YOLO, SimCLR, OpenCV, ONNX Deployment
Biomedical Analysis	Multimodal Signal Processing (PPG/ECG), Time-series Modeling
Software & Tools	Python, MATLAB, SOLIDWORKS, Java, Git, Flask, React
Languages	English (CET-4: 627, CET-6: 536), Chinese (Native)

AWARDS & HONORS

Dean's Honor Scholarship / Ruoshui Scholarship	2024 – 2025
Sole recipient in Zhiren College (968 students).	
SUSTech First-Class Merit Scholarship	2024 – 2025
Ranked 3/257 in comprehensive evaluation (Top 1.2%).	
SUSTech Outstanding Student Leader Pacesetter	2024 – 2025
Top 10 students university-wide.	

LEADERSHIP & SERVICE

President, Zhiren Student Development Committee	2025 – Present
Led student development and academic support initiatives at the college level; organization recognized as Outstanding Student Organization (First Prize).	
President, Zhiren Chinese Orchestra	2024 – 2025
Coordinated large-scale performances and organizational planning; organization recognized as Outstanding Student Organization, and received Outstanding Student Leader.	
Student Counselor, Zhiren College	2024 – 2025
Mentored 200+ freshmen in academic planning and campus adaptation.	